

Autonomous Pedagogical Agent for Grading and Explanatory Feedback: A Comparative Analysis of ReAct and Fine-Tuning Paradigms

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Abstract

This proposal presents an **Autonomous Pedagogical Agent (APA)** for assignment grading, named *AutoGrade+*. The agent generates scores and human-like explanations for assignments by combining Large Language Model (LLM) reasoning with structured, rubric-aware scoring logic. The core objective is to move beyond simple generative scoring by implementing a **Reasoning and Acting (ReAct)** framework, enabling the agent to use the rubric as a verifiable external tool [10]. Critically, this project performs a comprehensive **Comparative Analysis** between two distinct generative AI paradigms: 1) the proposed **ReAct Agentic** approach, and 2) a specialized **Low-Rank Adaptation (LoRA) Fine-Tuned** approach [3]. We will evaluate both methods on accuracy, explanation quality, and computational efficiency using academic evaluation datasets [1].

1 Introduction

Timely and consistent feedback is crucial for student learning, yet manual grading remains labor-intensive and susceptible to subjectivity. Automated Essay Scoring (AES) has evolved significantly [5], and recent advances with Large Language Models (LLMs) enable systems that not only predict scores but also generate detailed, constructive feedback [1]. This potential has driven a surge in educational AI research [8, 9].

This project addresses the need for a sophisticated, **Agentic AI system** capable of high-fidelity evaluation and providing human-quality explanations. We propose to develop an Autonomous Pedagogical Agent (APA) that leverages the **Reasoning and Action (ReAct) paradigm** [10] and structured tool-calling mechanisms to evaluate student submissions against an instructor-defined rubric. The primary research objective is a rigorous **comparative analysis** of the agentic and fine-tuning paradigms, satisfying the course's research requirement.

2 Literature Review

The theoretical foundation of this project rests on three pillars:

Agentic AI and Tool Use The ReAct framework [10] is the core architectural choice, as it allows the LLM to interleave reasoning (*Thought*) with external actions (*Action*), such as querying a knowledge base or using a tool. This mechanism, sometimes enhanced by Retrieval-Augmented Generation (RAG) [4], is essential for grounding the grading process in the specific, factual criteria of the rubric, moving beyond reliance on the LLM’s potentially generalized internal knowledge.

Automated Grading Research in Automated Essay Scoring (AES) has progressed from statistical models to efficient transformer-based architectures [5, 6]. Modern work extends these techniques to complex tasks, such as generating feedback for programming assignments [7]. The current state-of-the-art focuses on generating both the score and the justification, often termed *ASAG+F* (Automatic Short Answer Grading with Feedback) [1].

Model Adaptation A key comparative element is the choice of model adaptation. We compare the prompt-centric approach of ReAct against the data-centric approach of fine-tuning. Techniques like LoRA (Low-Rank Adaptation) [3] and Instruction Fine-Tuning [2] allow for efficient specialization of foundation models, providing a strong baseline for the performance comparison.

3 Proposed System: AutoGrade+ Agent Architecture

The *AutoGrade+* system implements a closed-loop ReAct architecture tailored for assessment.

3.1 System Components

1. **Rubric Tool (The Action):** A structured parser that converts the instructor’s rubric (expected answer, keywords, point values) into a programmatic tool. The agent’s action will be to query this tool with a student’s answer segment to receive an objective score component.
2. **LLM Core (The Thought):** A powerful, instruction-tuned LLM [2] serves as the reasoning engine. It decomposes the assignment, determines which part of the student answer maps to which rubric criterion, and plans the sequence of tool calls.
3. **ReAct Orchestrator:** Manages the loop: *Thought* $\rightarrow$ *Action* (Rubric Tool Call) $\rightarrow$ *Observation* (Tool Result) $\rightarrow$ new *Thought* $\rightarrow$ Final Score and Explanation.

4 Comparative Analysis (Research Focus)

The academic core of this project is the comparison between two dominant approaches to LLM specialization:

4.1 Methods Under Comparison

1. **Method A: ReAct Agentic Grading (Proposed):** Uses a pre-trained open-source LLM (e.g., Llama 3) in a few-shot ReAct loop with the Rubric Tool. This tests the performance of general-purpose LLMs when granted structured agency.

2. **Method B: LoRA Fine-Tuned Seq2Seq Model:** Uses a specialized, smaller model (e.g., Flan-T5) fine-tuned with LoRA [3] on a dataset of [Student Answer, Rubric, Score, Explanation] triplets. This model performs the task end-to-end without explicit tool calls.

4.2 Evaluation Metrics

Evaluation will use a curated set of short-answer/essay data and be quantified by:

- **Scoring Accuracy:** Mean Absolute Error (MAE) and Weighted Kappa agreement with human "gold standard" scores.
- **Fidelity:** Precision and Recall of the agent in correctly citing the specific rubric element used for scoring (testing transparency and verifiability).
- **Feedback Quality:** Human-judged assessment of the generated explanations for clarity, constructive nature, and relevance to the student's errors [1].
- **Efficiency:** Comparison of inference latency (speed) and token usage (cost) between the multi-step ReAct agent and the single-step fine-tuned model.

5 Conclusion

The *AutoGrade+* project is a research-level demonstration of Generative AI's application in education. By implementing and rigorously comparing the flexible ReAct agent paradigm against the efficient fine-tuned model baseline, the project will satisfy all course requirements while yielding valuable insights into the trade-offs between zero-shot agentic design and specialized parameter-efficient fine-tuning for automated pedagogical tasks.

References

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